

DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

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ABSTRACT

Parametric imitation learning via behavior cloning (BC) is fundamentally constrained by compounding error: during closed-loop deployment, small action errors accumulate and drive the agent into out-of-distribution states, where a single globally fit policy generalizes poorly. We show that this instability can be substantially mitigated—without departing from the standard behavior cloning setting—by retaining the expert demonstrations at inference time and recasting imitation learning as a semi-parametric, retrieval-based problem. We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, which reparameterizes the policy around local neighborhood structure rather than a single global state-to-action mapping. For each query state, DARP retrieves its k nearest neighbors in the expert dataset, forms one candidate action per neighbor from the neighbor state, its associated expert action, and the relative difference vector to the query, and aggregates these neighbor-conditioned candidates through a permutation-invariant operator to produce a single, robust prediction. Crucially, DARP operates entirely within the assumptions of behavior cloning: it introduces no additional assumptions, and requires no additional data collection, no online expert feedback, and no task-specific knowledge. We argue, through complementary analysis and experiment, that grounding each prediction in retrieved neighbors while enforcing local consistency reduces estimator variance and improves robustness to covariate shift. Across continuous control (MuJoCo), simulated and real robotic manipulation (Robosuite, RoboCasa), high-dimensional visual observations, and multimodal action distributions, DARP delivers consistent improvements of 15–46% over standard behavior cloning. Code and demos will be released upon acceptance.

1 INTRODUCTION

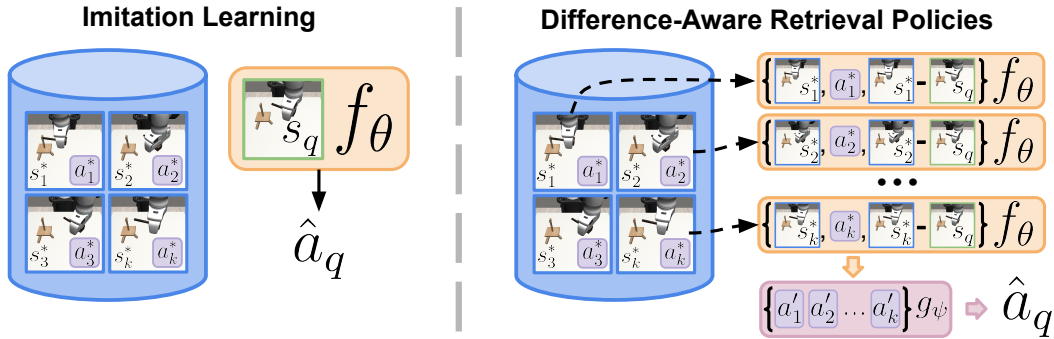


Figure 1: **Overview of DARP.** Standard BC (left) consumes the expert dataset only during training and, at deployment, predicts an action from the query state alone through a single global mapping. DARP (right) instead *retains* the dataset of expert state–action pairs and, at inference time, retrieves a local neighborhood of expert pairs around each query state s_q ; it then reparameterizes the policy around the difference vectors between the query and its retrieved neighbors, aggregating the resulting neighbor-conditioned candidate actions into a single prediction.

Imitation learning via behavior cloning (BC) (Pomerleau, 1991) has become a workhorse for endowing robots with complex, dexterous behaviors learned directly from expert demonstrations (Zhao et al., 2023; Chi et al., 2024; Black et al., 2024; Chung et al., 2014). Its appeal lies in its simplicity; that same simplicity, however, renders it brittle in practice, especially over long horizons (Ross et al., 2011). The governing failure mode is *covariate shift*: small prediction errors accumulate over a rollout and progressively carry the agent into states that the demonstration data represents only sparsely (Spencer et al., 2021; Ross et al., 2011). It is precisely in these out-of-distribution regions—where reliable control matters most—that BC policies become unstable, exhibiting high-variance behavior that frequently culminates in task failure.

Although this failure mode is widely recognized, and a substantial body of work has sought to curb compounding error (Ross et al., 2011; Venkatraman et al., 2015; Ke et al., 2024b; Levine et al., 2020), the prevailing remedies step outside the standard BC setting, presupposing access to simulators, interactive experts, large volumes of sub-optimal data, or strong task-specific structure. We deliberately retain the pure BC regime—learning solely from expert state–action pairs, with no additional supervision or feedback—and pose a correspondingly sharp question: *can the variance of a BC policy be reduced using nothing beyond the original demonstration dataset?* We answer this question affirmatively.

A statistical reading of BC locates the source of the difficulty precisely. Behavior cloning minimizes only the supervised risk on expert states; this controls bias on the training distribution but leaves the estimator’s variance essentially unconstrained, so that in the low-density regions routinely visited during closed-loop rollouts the learned policy is free to oscillate arbitrarily. The natural corrective is to impose *smoothness*—requiring nearby states to induce similar predicted actions (Kobayashi, 2022; Asadi et al., 2018; Ke et al., 2024a; Chen et al., 2024)—thereby suppressing spurious fluctuations and stabilizing rollouts. A range of mechanisms for encouraging such smoothness has been studied, which we survey in Section 4.

Each of these mechanisms, though effective in its intended setting, carries a characteristic limitation: data augmentation offers no guarantee of consistency, global priors tend to blur behaviorally distinct modes, temporal penalties regularize only along the time axis rather than across the state space, and explicit graph regularizers introduce additional smoothness hyperparameters that must be tuned.

A complementary axis of prior work contrasts global with local learning. “Global” supervised models (Black et al., 2024; Zhao et al., 2023; Chi et al., 2024) compress the entire demonstration dataset into a single parametric function and are, in consequence, brittle under distribution shift. “Local” methods (Pari et al., 2022; Mansimov & Cho, 2018; Salzberg & Aha, 1994) instead condition each prediction on the structure of the dataset itself, consulting neighborhoods of similar states and generating outputs through non-parametric or semi-parametric operations over the empirical distribution of expert behavior. Because locality avoids committing to a single global function, it confers robustness; yet it carries its own constraints, in that its behavior is dictated by the choice of distance metric, naive averaging over a neighborhood blurs distinct actions and cannot represent multimodality, and reasoning about neighbors solely through their absolute states forgoes available generalization.

We resolve this tension with **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, which unifies the robustness of local methods with the stability of regularized global policy learning within a single architecture. Rather than mapping the current query state to an action through one feedforward pass over a global parametric function, DARP (Fig. 1) retrieves k neighbors from the training corpus at inference time and produces k candidate actions, each conditioned on a tuple comprising the neighbor state, its associated expert action, and the difference vector from the query state; a permutation-invariant aggregation of these neighbor-conditioned candidates then yields a single, robust action. This construction simultaneously grounds every prediction in observed data, through non-parametric retrieval, and enforces local consistency, through parametric action prediction conditioned on the retrieved neighbors. It thereby reduces variance while remaining strictly within the assumptions of standard behavior cloning, requiring no additional data, no online supervision, and no task-specific knowledge. Viewed spectrally, and in its parameter-free averaging form, this neighbor aggregation approximates a Laplacian filter on the k -nearest-neighbor (k -NN) graph of expert states—a smoothing operator that adapts automatically to the local density and geometry of the dataset without any tuning.

We substantiate DARP through mutually reinforcing analysis and experiment. On the analytical side, we study an idealized, parameter-free instantiation of neighbor aggregation and prove that it applies a fixed low-pass spectral filter on the k -NN graph, formally equivalent to Laplacian regularization and provably reducing estimator variance while improving smoothness and closed-loop stability relative to standard BC; the practical DARP model, which conditions on difference vectors and admits learned aggregation, is designed to inherit this inductive bias, and our experiments test whether it does. Empirically, DARP attains 15–46% gains over standard behavior cloning across continuous control (MuJoCo), simulated robotic manipulation (Robosuite, RoboCasa), high-dimensional visual imitation (Robosuite with R3M image embeddings), a multimodal manipulation task, and a real-robot assembly task. These results establish that DARP is a general architecture that extends naturally to image-based observations and to expressive policy classes such as Set Transformers, Gaussian mixture models, and diffusion policies. A careful suite of ablations then isolates the contribution of each design choice, identifying the difference-based representation and permutation-invariant aggregation as the primary drivers of performance and yielding broadly applicable insight into retrieval-based algorithms for sequential decision-making.

2 DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

In this section we develop a class of imitation learning methods that combines the complementary strengths of “global” parametric learning and “local” neighbor-based learning. We introduce an architecture together with a deliberately simple training objective that learns under precisely the requirements of standard behavior cloning, yet, as we establish analytically and then evaluate empirically, yields substantial improvements in variance and robustness. We derive the accompanying guarantees from first principles; readers interested primarily in the practical algorithm may proceed directly to the self-contained summary of DARP in Section 2.5 and then to the empirical evaluation in Section 3.

The development proceeds in three stages. We first fix the problem setting (Section 2.1), then analyze a regularized variant of imitation learning (Section 2.2) that injects the geometric structure of the data into the objective to improve variance, generalization, and stability. Section 2.3 shows that these same benefits can be obtained *implicitly*, by modifying the policy *architecture* rather than the objective, in an idealized parameter-free setting that makes the equivalence exact. Finally, Section 2.4 instantiates the practical algorithm DARP, which realizes these benefits through a semi-parametric, retrieval-augmented architecture that applies broadly to imitation learning with modern neural networks and generative modeling tools.

2.1 PRELIMINARIES: BEHAVIOR CLONING FOR IMITATION LEARNING

We adopt the standard imitation learning setting, formalized as a finite-horizon Markov Decision Process (MDP) $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, P_0\}$ with state space $\mathcal{S}$, action space $\mathcal{A}$, and initial state distribution P_0 . A policy $f_\theta : \mathcal{S} \rightarrow \Delta_{\mathcal{A}}$ maps each state to a distribution over actions so as to maximize task-relevant objectives; for brevity we suppress the subscript θ when referring to the general function class f . We assume access to a fixed set of expert demonstrations $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ given as state–action pairs, and reserve the notation s^* and a^* for states and actions drawn from this expert dataset. Behavior cloning (Pomerleau, 1991) recovers a policy f_θ by casting imitation as supervised maximum-likelihood estimation, $\arg \max_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\log(f_\theta(a^* | s^*))]$. Although f_θ may in principle instantiate an arbitrarily expressive generative model (Lipman et al., 2023; Chi et al., 2024), we begin with a Gaussian parameterization for analytical clarity¹ and generalize to richer policy classes in the practical algorithm.

2.2 WARM-UP: NEIGHBOR MANIFOLD REGULARIZED IMITATION LEARNING

Because behavior cloning minimizes the supervised imitation loss only over expert states drawn from $\mathcal{D}^*$, it exercises no control over the policy’s behavior on states lying off the manifold of expert states. In practice, accumulating errors carry the agent into precisely these out-of-distribution regions, where

¹We relax this assumption in Section 2.4.

an unconstrained BC policy—and, most acutely, an overparameterized neural network—may act arbitrarily (Ross & Bagnell, 2010).

The root cause is that behavior cloning constrains function values only at isolated training states, never accounting for the relationships among states and their actions within a neighborhood, and thereby discards the geometry of the underlying data manifold. To restore this information, we augment the objective with a regularization term that explicitly enforces *local consistency*: states that are nearby in the expert dataset should map to similar actions. This principle yields the neighborhood-regularized loss $\mathcal{L}_{\text{MRIL}}$, which couples the standard imitation objective ($\mathcal{L}_{\text{BC}}$) with a smoothness penalty ($\mathcal{L}_{\text{S}}$) that compels predictions to respect the geometry of the dataset rather than relying on pointwise supervision alone.

$$\mathcal{L}_{\text{MRIL}}(f) = \underbrace{\mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)]}_{\text{supervised risk}(\mathcal{L}_{\text{BC}})} + \lambda \underbrace{\mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right]}_{\text{smoothness regularizer}(\mathcal{L}_{\text{S}})}, \quad (1)$$

where $\ell(f(s^*), a^*)$ is the supervised imitation loss, $\mathcal{N}_k(s^*)$ denotes the k -nearest neighbors of s^* within the expert dataset, and the weights $w_i(s^*)$ are normalized kernel weights determined by the state differences, $w_i(s^*) \propto K_{\Delta} \left(\frac{\|s_i^* - s^*\|}{h} \right)$. As sketched below and established in full in Appendix A.1.1, this penalty is precisely a form of *manifold regularization*, or Laplacian smoothing, that penalizes high-frequency variation of the policy across the neighborhood of expert states, and it provably improves the variance, stability, and generalization of the learned policy.

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.
- (ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.
- (iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^2$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Under these assumptions, therefore, MRIL enjoys strictly stronger generalization and stability guarantees than vanilla BC.

² L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

Proof sketch. The full argument appears in Appendix A.1.1; we sketch its structure here. The proof proceeds in three steps. First, we identify the smoothness regularizer with a graph Laplacian penalty on the graph whose k -nearest-neighbor (k -NN) affinity matrix is defined by the kernel weights w_i . Second, we show that, as the number of samples grows without bound, this graph Laplacian penalty converges to the weighted Dirichlet energy (Belkin & Niyogi, 2008; Zhou et al., 2003). Minimizing the Dirichlet energy then yields two consequences: it renders the learned f locally Lipschitz almost everywhere, which delivers smoothness and hence policy stability, and it acts as Tikhonov regularization, reducing estimator variance while keeping the bias controlled. $\square$

The regularizer thus does more than penalize pairwise disagreement between neighbors: it drives the learned policy toward smoothness with respect to the underlying data manifold. Concretely, it contracts the local Lipschitz constant of f along directions of high data density $p(s)$, so that small perturbations in state induce small, consistent changes in the predicted action. The resulting policy generalizes more reliably on in-distribution (ID) states and extrapolates in a structured, geometry-respecting manner to nearby out-of-distribution (OOD) states—exactly the regime into which compounding error drives a deployed policy.

2.3 IMPLICIT MANIFOLD REGULARIZATION VIA IN-CONTEXT ARCHITECTURES

Although the MRIL objective amortizes local learning and improves over vanilla BC, it carries two drawbacks. First, it introduces a hyperparameter λ that must be tuned to balance supervised accuracy against smoothness. Second, replacing the standard BC objective with a modified, regularized one can reshape the optimization landscape in adverse ways. Together these drawbacks motivate a sharper question: *can the benefits of MRIL (Eq. 1) be recovered by modifying the policy architecture rather than the objective?*

We answer this question affirmatively for an idealized, parameter-free instantiation. In this section we introduce a retrieval-based architectural modification that induces an *implicit* manifold regularization effect (iMRIL) while training under an entirely standard imitation objective. As Fig. 2 illustrates, iMRIL recovers the Laplacian smoothing of MRIL without introducing λ as a training hyperparameter; Section 2.4 then builds on this construction to develop the practical algorithm DARP, whose learned, difference-conditioned predictor is designed to inherit the same inductive bias.

iMRIL architecture: The core idea of iMRIL is to relocate the neighborhood aggregation (averaging) operation from the objective, where it appears in Eq. 1, into the architecture itself. Rather than training a standard feedforward predictor $f(s)$ against the neighborhood-regularized smoothness objective (Eq. 1), we embed the neighborhood-aggregation structure directly into the parameterization of the action predictor $\hat{f}$ while leaving the standard imitation objective unchanged. Concretely, iMRIL learns the parameters of a per-state predictor f_θ such that the aggregated predictor $\hat{f}(s^*) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*)$, formed by averaging f_θ over the nearest-neighbor states $\{s_i^*\}_{i \in \mathcal{N}_k(s^*)}$ drawn from the training set, accurately predicts the corresponding expert action a^* . Under this parameterization, iMRIL optimizes, at training time,

$$\arg \min_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} \left[\left\| \underbrace{\left(\frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*) \right)}_{\hat{f}(s^*)} - a^* \right\|_2 \right] \quad (2)$$

At deployment, inference on a new state s_q reduces to retrieving the k -NN of s_q from the training set and applying the same neighborhood aggregation, $\hat{a} = \hat{f}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*)$. As Section 2.4 shows, the specific parameterization of f is decisive for empirical performance, and it is precisely this consideration that leads to DARP.

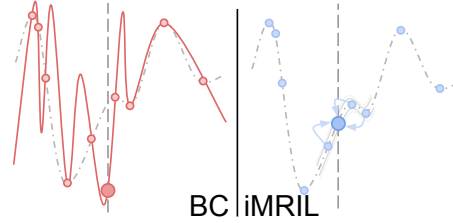


Figure 2: **iMRIL implicitly achieves Laplacian smoothing, reducing variance and enforcing local consistency**, whereas standard BC, unconstrained by any smoothness penalty, admits arbitrarily jagged interpolants between training states.

Intuitively, we parameterize the action predictor $\hat{f}$ as an aggregation of neighbor-state predictions $f(s_i^*)$ drawn from the training data, and then learn f . Supervising the *post-aggregation* function implicitly forbids any component prediction f from being arbitrarily non-smooth, thereby conferring the benefits established in Section 2.2. We make this correspondence exact for the parameter-free averaging case by proving a direct equivalence between iMRIL and the Laplacian regularization of Section 2.2.

Equivalence between iMRIL and MRIL: We defer the full proof of formal equivalence between MRIL and iMRIL to Appendix Section A.1, and state the main result together with a proof sketch here.

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL’s neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency ones. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} establishes that iMRIL is equivalent to a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof sketch. Full details appear in Appendix Section A.1.2. Diagonalizing the k -NN Laplacian gives the explicit regularizer a closed-form spectral solution, applying the filter $(1 + \lambda \mu)^{-1}$ to each eigenmode. The implicit objective, in turn, can be written as neighbor aggregation $\hat{f} = S f$ with $S = D^{-1} A$ the random-walk matrix, which shares the same eigenvectors and applies the fixed filter $1 - \mu$. Both are therefore low-pass filters on the graph: modes with small eigenvalues, corresponding to smooth variation across the data manifold, are largely preserved, whereas modes with large eigenvalues, corresponding to rapid, high-variance fluctuations between neighbors, are strongly damped. Consequently iMRIL performs Laplacian smoothing implicitly, reducing variance and enforcing local consistency with no need to tune λ . $\square$

We stress that this implicit-smoothing view does not obviate the need to learn a policy; it instead constrains the function class that can be represented *after* aggregation. The neighbor-conditioned network f_θ learns how expert actions vary under local perturbations, proposing a locally adapted action for each neighbor, while the aggregation operator smooths these proposals across the neighborhood to reduce variance. The result is a clean division of labor: learning supplies accuracy by correcting local bias, and aggregation supplies stability by controlling variance. This division of labor, established here for the idealized averaging operator, is the design principle that the practical algorithm of Section 2.4 carries over to learned predictors and aggregators.

2.4 DIFFERENCE-AWARE RETRIEVAL POLICIES: A PRACTICAL INSTANTIATION OF iMRIL FOR IMITATION LEARNING

Building on the conceptual framework of iMRIL, we now instantiate a practical algorithm for large-scale imitation learning. Starting from the objective in Eq. 2, we specify two design choices—(1) the parameterization of the predictor and (2) the neighbor aggregation operator—that together drive the strong empirical performance reported in Section 3.

2.4.1 DIFFERENCE-BASED PARAMETERIZATION OF f_θ

The objective in Eq. 2 leaves both the parameterization and the input representation of f_θ unspecified. Our guiding observation is that neighborhood aggregation should capture how expert actions vary

under local perturbations of the state, which implies that f_θ ought to reason explicitly about the *difference* between a query state and each neighbor state so as to propose a locally adapted action per neighbor. Accordingly, in **Difference-Aware Retrieval Policies (DARP)** we do not parameterize the predictor merely as $f_\theta(s_i^*)$; we instead supply it with additional context—the neighbor’s expert action a_i^* and the difference vector $\Delta s_i = s_i^* - s_q$ between the query and neighbor states. For a query state s_q and a neighbor (s_i^*, a_i^*) , the predictor f_θ then emits a candidate action $a'_i = f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)$ conditioned on this difference information. This difference-based conditioning is the distinguishing feature of DARP, and the ablations of Section 3 identify it as the single most important contributor to performance.

Let $\mathcal{N}_k(s_q)$ denote the index set of the k -nearest neighbors retrieved under a distance function $d(s_q, s_i^*)$.³ To generate a prediction for any query state s_q , DARP applies the neighborhood aggregation of Section 2.3:

$$\hat{a}_q = f_{\text{DARP}}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i \quad (3)$$

$$= \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q). \quad (4)$$

This prediction rule induces a straightforward imitation learning objective over the expert dataset $\mathcal{D}^*$ at training time:

$$\arg \min_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\|f_{\text{DARP}}(s_q^*) - a_q^*\|^2] \quad (5)$$

which optimizes the parameters of the predictor f_θ so as to minimize the discrepancy between the aggregated prediction $\hat{a}_q$ and the expert action a_q^* . Because the objective is a plain regression loss, f_θ admits essentially any parameterization; we use standard feedforward or convolutional neural networks. As Section 3 confirms, this difference-based parameterization is decisive for performance. At inference time, actions are generated by retrieving the k -NN of the query and evaluating the neighborhood aggregation of Eq. 3.

2.4.2 GOING BEYOND LINEAR AGGREGATION

The neighborhood aggregation considered so far—the simple average $\hat{a}_q = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i$ —is but one instance of a far broader class of permutation-invariant aggregation functions $g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q)})$. In place of averaging, g_ψ can be realized by expressive set-equivariant neural models such as the Set Transformer (Lee et al., 2019) or DeepSets (Zaheer et al., 2017), generalizing the prediction rule of Eq. 3 to $\hat{a}_q = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$. Beyond added expressivity, replacing the fixed average with a parametric aggregator g_ψ enables DARP to represent action distributions substantially richer than the unimodal Gaussian implicit in the L_2 -regression objective of Eq. 5—for instance Gaussian mixture models (Pignat & Calinon, 2019) or diffusion models (Chi et al., 2024). Rather than predicting $\hat{a}_q$ directly, DARP can instead predict the parameters α of an action distribution $p(a_q; \alpha)$ —such as the means, covariances, and mixture weights of a Gaussian mixture model, or the score function of a diffusion model—and thereby perform maximum-likelihood training of multimodal action distributions in place of unimodal L_2 regression:

$$\arg \max_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\log p(a_q^*; \alpha_\theta(s_q^*))], \text{ where } \alpha_\theta(s_q^*) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q^*)\}_{i \in \mathcal{N}_k(s_q^*)}) \quad (6)$$

Inference for a query state s_q then proceeds by first constructing $\alpha_\theta(s_q) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$ from the neighbors retrieved at test time, and then sampling $\hat{a}_q \sim p(\cdot; \alpha_\theta(s_q))$. This modularity—an arbitrary predictor f_θ composed with an arbitrary permutation-invariant head g_ψ —is what allows DARP to serve as a drop-in enhancement across MLP, Set Transformer, GMM, and diffusion backbones, as the experiments confirm. Detailed training pseudocode is provided in Appendix Section A.3.

³We use Euclidean distance in a pre-trained embedding space, though other neighborhood functions apply equally well; Appendix Section A.2.1 discusses the design decisions involved in constructing neighborhood sets via retrieval, including the near-identical performance obtained under Euclidean and cosine metrics.

2.5 SUMMARY

To summarize, we have established that enforcing local consistency when fitting a behavior cloning policy amounts to manifold regularization (Laplacian smoothing), and provably improves variance, smoothness, and stability relative to standard BC (Theorem 1). We have further shown that, in the parameter-free averaging setting, this smoothing need not be imposed through an explicit regularization term and its attendant hyperparameter λ : it can be induced implicitly by building neighborhood aggregation into the policy architecture itself (Theorem 2). These principles crystallize into a single algorithm. At training time, for each expert state-action pair (s_q^*, a_q^*) , DARP retrieves the k -nearest-neighbor states from $\mathcal{D}^*$, forms the difference vectors $\Delta s_i = s_i^* - s_q^*$ from each neighbor to the query, passes each neighbor tuple $(s_i^*, a_i^*, \Delta s_i)$ through the network f_θ to obtain candidate actions, and aggregates these candidates with a permutation-invariant function g_ψ to produce the final prediction—all trained under the standard imitation learning objective alone. The remainder of the paper subjects each of these choices to empirical scrutiny.

3 EXPERIMENTAL EVALUATION

We now evaluate DARP, organizing the study around three questions: **Q1**: Does DARP consistently outperform standard behavior cloning? **Q2**: Does DARP accommodate more complex state representations and action distributions? **Q3**: Which architectural components drive DARP’s performance gains? To address these questions comprehensively, we evaluate across multiple domains and across low-dimensional states, high-dimensional image features, and diverse action representations. The evaluation spans continuous control (MuJoCo), simulated robotic manipulation (Robosuite and RoboCasa), high-dimensional visual observations, multimodal action distributions, a real-robot assembly task, and deliberately discontinuous environments constructed to stress-test the neighbor-based approach.

3.1 BASELINE COMPARISONS AND TASK DESCRIPTIONS

MuJoCo Tasks: The MuJoCo (Todorov et al., 2012; Fu et al., 2020) tasks require controlling legged figures of several distinct embodiments to achieve forward locomotion on a flat plane, spanning Hopper (a single-legged hopping robot), Walker (a bipedal humanoid), Ant (a quadruped), and HalfCheetah (a biped).

Robosuite Tasks: The Robosuite (Zhu et al., 2020) tasks each require a single robotic arm to manipulate objects. The Stack task requires placing a smaller cube atop a larger one; the Threading task requires manipulating a thin, needle-like tool and inserting it into a small ring; and the Square Peg task requires grasping a square wooden block with a central hole and seating it onto a square peg.

RoboCasa Tasks: The RoboCasa (Nasiriany et al., 2024) tasks each require a single robotic arm to manipulate objects within a randomized kitchen setting. The Drawer task requires closing an open drawer, the Door task requires closing an open microwave oven door, and the Stove task requires twisting a knob to turn off a stove burner.

Baseline Comparisons: We benchmark DARP against a set of baselines and ablations spanning the non-parametric, parametric, and in-context learning paradigms: (1) **R&P** (Sridhar et al., 2025), which directly executes the action of the single nearest neighbor; (2) **LWR** (Pari et al., 2022), which performs locally weighted regression over the retrieved neighbors; (3) **BC**, standard parametric behavior cloning; (4) **REGENT** (Sridhar et al., 2025), a transformer-based in-context learning method conditioned on retrieved neighbors; and (5) **MRIL**, the explicitly smoothed variant of DARP introduced in Section 2.2. This suite situates DARP against representatives of every paradigm to which it is related—pure retrieval, local regression, global parametric learning, in-context conditioning, and explicit smoothing—so that its gains can be attributed to the unification these baselines individually lack. All parametric policies use matched MLP backbones unless stated otherwise; a comparison against a diffusion-policy backbone appears in Appendix Table 8.

3.2 CAN DARP CONSISTENTLY OUTPERFORM STANDARD BEHAVIOR CLONING? (Q1)

We first test DARP’s central hypothesis on tasks with low-dimensional state representations, where inter-state distances are well-defined and interpretable. This evaluation spans locomotion tasks

from MuJoCo and robotic manipulation tasks from Robosuite and RoboCasa, with manipulation demonstration data generated by MimicGen (Mandlekar et al., 2023). Throughout these experiments the aggregation function g is the simplest possible instantiation—a uniform average over all neighbor action predictions a' —so that the gains reported below cannot be attributed to a more expressive aggregator, and instead isolate the contribution of difference-based retrieval alone.

Method	Hopper	Ant	Walker	HalfCheetah
R&P (Sridhar et al., 2025)	711.82 $\pm$ 85.63	-305.97 $\pm$ 76.42	419.18 $\pm$ 50.21	-178.64 $\pm$ 29.75
LWR (Pari et al., 2022)	1703.78 $\pm$ 245.95	846.59 $\pm$ 216.06	1484.91 $\pm$ 356.54	1945.82 $\pm$ 567.26
BC	2313.65 $\pm$ 203.75	2376.20 $\pm$ 339.43	2658.40 $\pm$ 274.08	1063.23 $\pm$ 371.08
REGENT (Sridhar et al., 2025)	1819.39 $\pm$ 186.24	-302.10 $\pm$ 146.67	507.01 $\pm$ 76.10	169.85 $\pm$ 63.10
MRIL	2793.63 $\pm$ 156.41	3869.08 $\pm$ 241.00	4370.96 $\pm$ 168.13	701.58 $\pm$ 195.08
DARP	3545.57 $\pm$ 3.54	4383.28 $\pm$ 266.37	4894.01 $\pm$ 75.12	5515.41 $\pm$ 841.33
DARP Set Transformer	2965.86 $\pm$ 103.08	4063.79 $\pm$ 218.80	4752.42 $\pm$ 109.23	3417.85 $\pm$ 764.57

Table 1: **DARP attains the highest return on all four MuJoCo locomotion tasks, and its Set Transformer variant surpasses every non-DARP baseline on each task.** Return of DARP against BC and the non-parametric (R&P, LWR), in-context (REGENT), and explicitly regularized (MRIL) baselines on MuJoCo environments using low-dimensional state. Each score is the mean return over 100 independent trials with 95% confidence intervals; the best result per task is in bold and is achieved by DARP in every case. The Set Transformer aggregator is a second, more expressive instantiation of DARP: on every task it exceeds all non-DARP baselines—including, on HalfCheetah, the BC baseline (3417.85 vs. 1063.23)—while the uniform-average DARP remains the top performer. Parametric policies use matched Multi-Layer Perceptron backbones; for results with diffusion policies, see Appendix Table 8.

The results establish that DARP improves consistently over standard behavior cloning across every tested environment. On the simulated manipulation tasks the gains span 15–31 percentage points in success rate (Table 2), and on the MuJoCo locomotion tasks the return improvements are pronounced—for example, DARP raises HalfCheetah return from BC’s 1063.23 to 5515.41 and Walker from 2658.40 to 4894.01 (Table 1). The comparison is equally informative about the baselines. The purely non-parametric methods (R&P and LWR) fare poorly across the board, confirming that retrieval alone is insufficient; the in-context REGENT baseline is unstable on the locomotion tasks; and while the explicitly regularized MRIL almost always exceeds vanilla BC, the highest score on every task in this suite is attained by the DARP architecture, with the Set Transformer variant likewise surpassing all non-DARP baselines. This pattern isolates the source of the gains: neither non-parametric retrieval nor explicit regularization alone accounts for them, and it is their unification within DARP—difference-based conditioning composed with permutation-invariant aggregation—that is responsible.

Having established these gains on low-dimensional states, we next ask whether the practical instantiation of Section 2.4 extends to substantially higher-dimensional input representations such as images.



Figure 3: The “DrawerClose” RoboCasa task: the robotic arm must close an open drawer within a randomized kitchen scene.



Figure 4: The “Threading” Robosuite task: the robotic arm must insert a needle-like implement into a small ring.

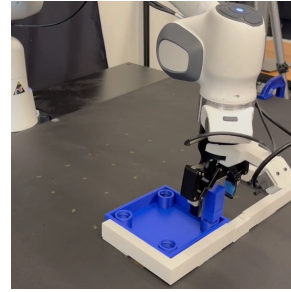


Figure 5: The real-world FurnitureBench square-table assembly task: the robot must pick up a table leg and screw it into a hole at the corner of the table-top.

Method	RoboCasa			Robosuite			Real
	Drawer	Door	Stove	Stack	Thrd.	Peg	Sq. Table
BC	54	29	28	47	37	46	44
DARP	85	45	43	72	63	62	92

Table 2: **DARP outperforms BC on all seven manipulation tasks across three evaluation environments, including a real robot.** Success rates (%) using low-dimensional state features (RoboCasa and Robosuite: 100 trials each; Real: 50 trials). DARP exceeds BC on every task, with margins of 15–31 points in simulation; the real-world square-table result, in which DARP roughly doubles BC’s success rate (92 vs. 44), is a single-task, single-baseline study and is best read as promising preliminary evidence of real-world robustness rather than a definitive demonstration. The RoboCasa and Robosuite policies use matched MLP backbones; the real-world policy uses a diffusion backbone (Appendix Table 8).

3.3 CAN DARP HANDLE MORE COMPLEX STATE REPRESENTATION AND ACTION DISTRIBUTIONS? (Q2)

High-Dimensional Visual Input Representations. To establish that DARP extends beyond the regime of compact, low-dimensional states, we evaluate it on the three Robosuite manipulation tasks with observations given by frozen, pretrained R3M image embeddings (Nair et al., 2022), testing whether neighbor retrieval remains effective when states are high-dimensional feature vectors extracted from raw visual input (Table 3). The representation is held fixed—neighbors are retrieved by Euclidean distance directly in the R3M embedding space—rather than learned jointly with DARP, so any gain is attributable to the policy rather than to representation tuning. Not only does DARP continue to outperform BC in this setting, but its average improvement of $\sim 35\%$ exceeds the $\sim 22\%$ average improvement observed on the low-dimensional Robosuite tasks. The margin between DARP and BC therefore *widens* as the state representation grows more complex, indicating that DARP exploits high-dimensional visual features more effectively than BC rather than merely tolerating them.

Method	Stack	Thrd.	Peg
BC	44	38	17
DARP	75	76	52

Table 3: **DARP scales to visual inputs**, outperforming BC on all three vision-based Robosuite tasks (success rates, %; states are frozen, pretrained R3M image embeddings, over 100 trials).

Multimodal Action Distributions. DARP also handles imitation tasks that demand genuinely multimodal action modeling: on the Push-T environment, equipped with a Set Transformer aggregator that predicts the parameters of a Gaussian mixture model, it improves over BC by more than 20%. This confirms that the permutation-invariant aggregator can be instantiated to represent multimodal action distributions, and that the choice of aggregation head is the sole modification required to do so. Full details are deferred to Appendix A.2.2.

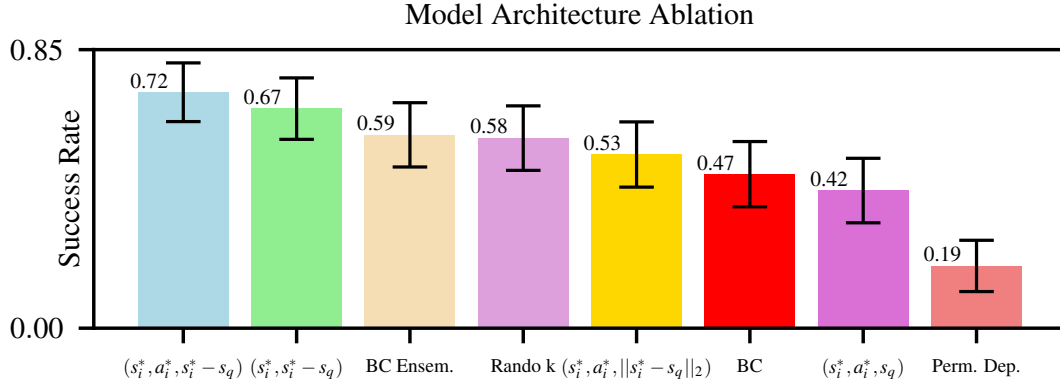


Figure 6: Difference vectors and permutation-invariant aggregation are the primary drivers of DARP’s success. Perturbing each core architectural component in turn shows that performance is most strongly attributable to the difference-based neighbor representation $(s_i^*, a_i^*, s_i^* - s_q)$, with permutation invariance a close second; removing the neighbor actions or the retrieval metric degrades performance more mildly. Success rate is averaged over 100 trials on the Robosuite Stack environment, with 95% confidence intervals.

3.4 HOW DO DIFFERENT ARCHITECTURAL COMPONENTS CONTRIBUTE TO DARP’S PERFORMANCE GAINS? (Q3)

Ablation Study: To isolate which components of the DARP architecture drive its gains, we conduct a systematic ablation that removes or alters each design choice in turn: (1) standard DARP; (2) DARP without the neighbor actions; (3) an ensemble of 10 BC agents; (4) DARP with randomly chosen neighbors in place of distance-based retrieval; (5) DARP using only the L_2 norm of the difference vector rather than its full direction; (6) the BC baseline, conditioned on the query state s_q alone; (7) DARP conditioned on the query state in place of the query-to-neighbor difference vector; and (8) DARP with a permutation-*dependent* (i.e. **not** permutation-invariant) aggregator over the a_i ’s. Together these variants disentangle the four ingredients of DARP—retrieval quality, difference conditioning, neighbor actions, and permutation invariance. Figure 6 reports the results.

The ablation yields a clear ordering of importance: the difference vectors and permutation invariance are the decisive contributors to DARP’s success, whereas the neighbor actions play a more modest role. Random neighbor selection collapses performance, confirming that a meaningful distance metric governing which neighbors are retrieved is essential—the difference representation cannot compensate for uninformative neighbors. Likewise, the permutation-invariant aggregator g is indispensable: permutation-*dependent* alternatives degrade performance markedly, precisely as the implicit-smoothing analysis predicts, since only a permutation-invariant operator realizes the neighbor-averaging (spectral filtering) that underlies the variance reduction.

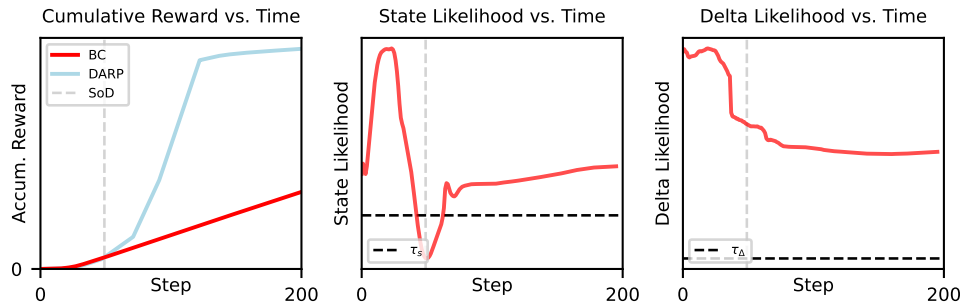


Figure 7: Cumulative reward for BC and DARP on the Robosuite Stack task: the rollouts are initially identical but diverge as BC fails and DARP succeeds. The vertical dashed line marks the step of divergence (“SoD”). At the SoD the query state is out-of-distribution (state likelihood $< \tau_s$) while its difference vector remains in-distribution (delta likelihood $> \tau_\Delta$)—precisely the regime in which the difference-based reparameterization is expected to help, and a direct empirical signature of the mechanism proposed in Section 2.4.

Divergence Analysis: To pinpoint the mechanism behind DARP’s advantage over standard BC, we analyze the point of divergence in rollouts where BC fails but DARP succeeds, defining the “step of

divergence” (SoD) as the step at which the two policies begin to receive significantly different reward. We set the thresholds τ_s and τ_Δ to the first percentile of training-set likelihoods, so that only 1% of the states, and respectively the difference vectors, observed at training time are less likely. Across all six rollouts spanning two tasks (Robosuite Stack and MuJoCo Hopper), the query state at the SoD is out-of-distribution in state likelihood ($< \tau_s$) yet remains in-distribution in difference-vector likelihood ($> \tau_\Delta$). This consistent signature directly corroborates our hypothesis that DARP’s gains arise, in part, from improved prediction on mildly out-of-distribution states—the very states into which compounding error carries a deployed policy—which the difference-vector reparameterization keeps within the support the network was trained on (Figure 7 plots the reward drift, SoD, and state and delta likelihoods for one representative task). Appendix A.2.4 reports complementary fork-and-recover experiments that further characterize DARP’s robustness to accumulated error.

4 RELATED WORK

Non-Parametric Imitation Learning Methods: Non-parametric IL algorithms attain strong performance by exploiting local structure in the demonstration data. VINN Pari et al. (2022) applies locally weighted regression to imitation and reports notably strong results in image-embedding spaces, while MiDiGaP von Hartz et al. (2025) models multimodal trajectories with mixtures of Gaussian processes to achieve rapid generalization. SEABO Lyu et al. (2024) repurposes retrieval for offline RL, constructing a reward function that rewards transitions lying close to their neighbors. FlowRetrieval Lin et al. (2024), STRAP Memmel et al. (2025), and Behavior Retrieval Du et al. (2023) couple non-parametric retrieval with finetuning over large unlabeled datasets, enabling generalization through test-time training. DARP is distinguished from all of these along two axes: its difference-based reparameterization, which encodes each retrieved neighbor as a $(s_i^*, a_i^*, s_i^* - s_q)$ tuple rather than by absolute state, and its genuinely semi-parametric policy, which learns from the retrieved neighborhood rather than relying on non-parametric aggregation or test-time finetuning alone. This combination is precisely what allows DARP to fuse the variance reduction of local methods with the generalization of parametric policies within one architecture trained under the standard BC objective.

Smoothness-Constrained Policy Learning: A substantial literature imposes explicit smoothness constraints to improve policy stability and robustness. L2C2 Kobayashi (2022) studies model-free RL under local Lipschitz-continuity constraints, attaining smoothness and noise robustness without sacrificing expressiveness, and Asadi et al. (2018) develop an analogous Lipschitz-constrained approach for model-based RL. CCIL Ke et al. (2024a) extends these ideas to imitation learning by generating synthetic corrective labels from a Lipschitz-constrained dynamics model, and the same principle has been scaled to humanoid controllers Chen et al. (2024) to reduce deployment-time shakiness. DARP departs from this line in a fundamental respect: it enforces smoothness *implicitly*, through an architectural change, while training under a standard imitation objective, thereby dispensing with the additional smoothness hyperparameters these methods require.

In-Context Learning Methods: A parallel line of work approaches retrieval through the lens of in-context imitation learning. REGENT Sridhar et al. (2025) pursues retrieval-augmented generalization by feeding retrieved states, actions, and rewards into a causal transformer, and DPT Lee et al. (2023) pretrains transformers by supervision to predict actions from a query state and an in-context dataset, effectively learning to explore. Related in-context architectures include ICRT Fu et al. (2024), Instant Policy Vosylius & Johns (2025), and KAT Di Palo & Johns (2024). Whereas these methods target rapid adaptation to new tasks and environments, DARP targets higher performance and stability on standard imitation learning, and its difference-based, permutation-invariant aggregation constitutes a distinct and complementary use of retrieval—a point made concrete by our head-to-head comparison against REGENT.

5 CONCLUSION

We introduced **D**ifference-Aware **R**etrieval **P**olicies for Imitation Learning (**DARP**) (DARP), a nearest-neighbor-based algorithm that reparameterizes imitation learning around the relative differences between query states and their nearest neighbors rather than learning a single global state-to-action mapping. For an idealized parameter-free instantiation, we proved that neighbor

aggregation implicitly realizes Laplacian smoothing, and that this smoothing provably reduces variance and improves closed-loop stability over standard behavior cloning—without leaving the standard BC regime or introducing a regularization hyperparameter—and we designed the practical, difference-conditioned DARP to inherit this inductive bias. Our evaluation across continuous control, simulated and real robotic manipulation, high-dimensional visual observations, and multimodal action distributions substantiates three claims. First, DARP consistently outperforms standard behavior cloning on low-dimensional state representations. Second, this advantage is preserved—and, for high-dimensional visual inputs, amplified—across different state representations, action-distribution modeling requirements, and task complexities, and across MLP, Set Transformer, GMM, and diffusion backbones, with improvements of 15–46% across the tested scenarios; the real-robot result, though limited to a single task, points to the same conclusion. Third, architectural ablations identify the difference vectors and permutation-invariant aggregation as the components responsible for these gains. Together, these results establish difference-aware retrieval as a simple, general, and drop-in mechanism for making behavior cloning markedly more robust to distribution shift, and they motivate future study of its behavior at very large dataset scales and in sharply discontinuous state spaces.

6 REPRODUCIBILITY STATEMENT

To support full reproducibility, we will release upon acceptance the supplementary source code, the policy and environment configuration files, the MuJoCo experiment data, the scripts that generate expert demonstrations for the Robosuite tasks via MimicGen, and the modality-transformation code. On NVIDIA L40 and L40s GPUs, the released artifacts reproduce the reported results exactly, with the sole exception of the transformer-based results (REGENT and the Set Transformer), whose stochastic training may induce slight deviations from the reported numbers.

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A APPENDIX

A.1 LEMMAS AND PROOFS

A.1.1 PROOF OF THEOREM 1

To prove Theorem 1, we begin from a well-established result (Chung, 1997; Zhou et al., 2003; Belkin & Niyogi, 2008).

Lemma 1 (Smoothness regularizer as k -NN graph Laplacian penalty). *Let $\{s_1^*, \dots, s_n^*\}$ be the expert states with corresponding predicted actions $f(s_i^*) \in \mathbb{R}^{d_a}$. For each i , let $\mathcal{N}_k(s_i^*)$ denote the indices of the k -nearest neighbors of s_i^* (excluding i). Define asymmetric weights*

$$\tilde{W}_{ij} = \begin{cases} w_j(s_i^*), & \text{if } j \in \mathcal{N}_k(s_i^*), \\ 0, & \text{otherwise,} \end{cases}$$

and construct a symmetric affinity matrix

$$W_{ij} = \frac{1}{2} (\tilde{W}_{ij} + \tilde{W}_{ji}).$$

Let D be the degree matrix with $D_{ii} = \sum_j W_{ij}$, and define the k -NN graph Laplacian $L = D - W$.

Then the smoothness regularizer can be written as the quadratic form

$$\mathcal{L}_S(f) = \frac{1}{n} \sum_{i=1}^n \sum_{j \in \mathcal{N}_k(s_i^*)} w_j(s_i^*) \|f(s_i^*) - f(s_j^*)\|^2 \propto \text{Tr}(F^\top L F),$$

where $F = [f(s_1^*), f(s_2^*), \dots, f(s_n^*)]^\top \in \mathbb{R}^{n \times d_a}$. Equivalently, in the scalar case,

$$\mathcal{L}_S(f) \propto f^\top L f.$$

Corollary 1 (Continuum limit of smoothness regularizer). *Assume states $\{s_i^*\}_{i=1}^n$ are sampled i.i.d. from a smooth density $p(s^*)$ supported on an m -dimensional C^2 manifold $\mathcal{M} \subset \mathbb{R}^d$. Let W be the symmetrized k -NN affinity matrix constructed from a kernel K_Δ with bandwidth h , and let $L = D - W$ be the graph Laplacian.*

If $n \rightarrow \infty$, $h \rightarrow 0$, and $nh^{m+2} \rightarrow \infty$, then the normalized quadratic form converges to the weighted Dirichlet energy:

$$\frac{1}{n^2 h^{m+2}} \text{Tr}(F^\top L F) \rightarrow C_K \int_{\mathcal{M}} \|\nabla_{\mathcal{M}} f(s)\|_2^2 p(s)^2 d\text{vol}(s),$$

where $C_K > 0$ is a constant depending only on the kernel K_Δ .

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.

(ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.

(iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^4$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Under these assumptions, therefore, MRIL enjoys strictly stronger generalization and stability guarantees than vanilla BC.

Proof. We prove each claim in turn.

(i) Variance reduction. The Laplacian penalty in $\mathcal{L}_{\text{MRIL}}$ is

$$\sum_{i,j} W_{ij} \|f(s_i^*) - f(s_j^*)\|^2 = 2f^\top L f,$$

where $L = D - W$ is the graph Laplacian, and W is the k -NN affinity matrix. By Lemma 1, this equals the empirical Dirichlet energy of f on the k -NN graph. It is well known (Zhou et al., 2003; Belkin & Niyogi, 2008) that such a quadratic penalty is equivalent to Tikhonov regularization with respect to the graph Laplacian norm $\|f\|_L^2 = f^\top L f$. In statistical learning theory, adding a Tikhonov penalty strictly reduces the variance of the estimator compared to the unregularized solution while keeping the bias term controlled. Thus MRIL enjoys smaller estimator variance than vanilla BC, which uses no such penalty.

(ii) Smoothness guarantee. Consider the continuum limit (Corollary 1): for i.i.d. samples $\{s_i^*\}$ from density p on a smooth manifold $\mathcal{M}$, the normalized penalty converges to

$$\int_{\mathcal{M}} \|\nabla f(s)\|^2 p(s)^2 d\text{vol}(s).$$

This is the weighted Dirichlet energy of f on $\mathcal{M}$. If this integral is finite, f belongs to the Sobolev space $H^1(\mathcal{M}, p^2)$, and in particular f is locally Lipschitz almost everywhere with

$$\|f(s^*) - f(s'^*)\| \leq C \|s^* - s'^*\| \quad \text{for } p\text{-a.e. neighbor pairs } s^*, s'^*.$$

Therefore minimizers of $\mathcal{L}_{\text{MRIL}}$ have uniformly bounded local Lipschitz constants along high-density regions of the state space. By contrast, minimizers of vanilla BC have no such constraint: any oscillatory interpolant that matches the training data exactly yields the same supervised risk, so arbitrarily large Lipschitz constants are possible.

(iii) Policy stability. Let $\Delta_t = \|s_t - s_t^*\|$ denote the deviation at time t . For Lipschitz dynamics T ,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|.$$

Decompose the action error:

$$\|f(s_t) - f^*(s_t^*)\| \leq \|f(s_t) - f^*(s_t)\| + \|f^*(s_t) - f^*(s_t^*)\|.$$

For vanilla BC, the first term $\|f(s_t) - f^*(s_t)\|$ is only minimized on the empirical distribution $P_{\mathcal{S}}$; off-distribution, it may be $O(1)$ regardless of Δ_t . The second term satisfies $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Hence the recursion can take the form

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(1) + O(\Delta_t)),$$

which accumulates linearly in t .

⁴ L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

For MRIL, the Laplacian penalty enforces

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for neighbor pairs } (s^*, s'^*),$$

as shown in part (ii). Thus $\|f(s_t) - f^*(s_t)\| = O(h^2)$ (where h is the radius of the kernel bandwidth around the point s_t) by local-linear regression error bounds, and $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Combining these,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a(O(h^2) + O(\Delta_t)).$$

Since the constant multiplying Δ_t is strictly smaller under the smoothness constraint, the cumulative error grows strictly slower than in vanilla BC. In particular, error growth is sublinear in the rollout horizon when h is small, whereas it is linear for vanilla BC.

Conclusion. Claims (i)–(iii) establish that MRIL yields lower variance, uniform smoothness control, and sublinear rollout error accumulation compared to vanilla BC, completing the proof. $\square$

Kernel choice. For the IC smoothness regularizer, we adopt a Gaussian kernel

$$w_i(s^*) \propto \exp\left(-\frac{\|s^* - s_i^*\|^2}{2h^2}\right), \quad \sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) = 1,$$

with bandwidth h set to the median distance to the k -th nearest neighbor across the dataset. This choice is standard in manifold regularization (Belkin & Niyogi, 2008; Zhou et al., 2003) and guarantees that the graph Laplacian penalty converges to the Dirichlet energy in the continuum limit. Empirically, this default proved stable across every task we considered, and alternative kernels (e.g. uniform k -NN or exponential decay) yielded qualitatively equivalent results, indicating that the smoothing effect is governed by the neighborhood geometry rather than by the precise kernel shape.

A.1.2 PROOF OF THEOREM 2

We first establish the two lemmas that characterize the spectral form of explicit Laplacian regularization and of neighbor aggregation, respectively.

Lemma 2 (Spectral form of explicit Laplacian regularization). *Let L be the symmetric normalized graph Laplacian with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$. The minimizer of the penalized objective*

$$\mathcal{L}_\lambda(f) = \|f - a^*\|^2 + \lambda f^\top L f$$

has the closed-form expansion

$$f_\lambda = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j.$$

Thus λ directly determines the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$ applied to each Laplacian mode.

Proof. Diagonalize $L = U\Lambda U^\top$ with $U = [u_1, \dots, u_n]$ orthogonal and $\Lambda = \text{diag}(\mu_1, \dots, \mu_n)$. Write $f = Uc$, $a^* = Ub$ in this basis. The objective becomes

$$\|Uc - Ub\|^2 + \lambda c^\top \Lambda c = \sum_{j=1}^n (c_j - b_j)^2 + \lambda \mu_j c_j^2.$$

Minimizing each term yields $c_j = \frac{1}{1 + \lambda \mu_j} b_j$. Transforming back gives the stated expansion. $\square$

Lemma 3 (Spectral form of neighbor aggregation). *Let $S = D^{-1}A$ be the random-walk matrix of the k -NN graph, with adjacency A and degree D . For any prediction vector f , the neighbor-averaged prediction is $\hat{f} = Sf$. In the Laplacian eigenbasis, this corresponds to the spectral filter*

$$\hat{f} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j,$$

i.e. $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

Proof. By definition, $L = I - D^{-1/2}AD^{-1/2}$ and $S = D^{-1}A = I - L_{\text{rw}}$ where $L_{\text{rw}} = D^{-1}L$ is the random-walk Laplacian. Since L_{rw} and L share the same spectrum up to similarity transform, the eigenbasis $\{u_j\}$ diagonalizes S . Thus for each mode u_j , $Su_j = (1 - \mu_j)u_j$, yielding the claimed spectral filter. $\square$

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency ones. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} establishes that iMRIL is equivalent to a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof. From Lemma 3, the neighbor aggregation operator $S = D^{-1}A$ acts on Laplacian eigenmodes u_j as

$$Su_j = (1 - \mu_j)u_j,$$

where μ_j are the normalized Laplacian eigenvalues. Thus in the graph Fourier basis, neighbor aggregation corresponds to multiplying each mode by the fixed spectral filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

On the other hand, Lemma 2 shows that explicit Laplacian regularization with parameter λ yields the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$. Both filters downweight high-frequency modes ($\mu \gg 0$) while preserving low-frequency modes ($\mu \approx 0$). The key difference is that $\phi_\lambda(\mu)$ requires tuning λ , whereas $\phi_{\text{DARP}}(\mu)$ is parameter-free.

To see the equivalence, note that for small μ ,

$$\phi_{\text{DARP}}(\mu) = 1 - \mu \approx (1 + \mu)^{-1} = \phi_{\lambda=1}(\mu) \quad \text{up to } O(\mu^2) \text{ terms.}$$

Thus DARP can be interpreted as performing Laplacian smoothing with an effective regularization weight of order $\lambda \approx 1$ in normalized units. Moreover, for large μ , $\phi_{\text{DARP}}(\mu)$ damps high-frequency modes even more strongly by driving them toward zero, providing a sharper low-pass effect than explicit regularization.

Therefore, DARP's aggregation step is mathematically equivalent to implicit Laplacian regularization with fixed spectral filter ϕ_{DARP} , eliminating the need to tune λ explicitly. $\square$

In its parameter-free averaging form, DARP can therefore be understood as a form of *locally adaptive implicit regularization*: in place of an explicit global weight λ , its neighbor aggregation step enforces smoothness automatically through the graph structure, so that the effective regularization strength varies with local degree and neighborhood geometry and adapts to the density of the expert demonstrations. Spectrally, this is the fixed filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$, which suppresses high-frequency modes more aggressively than any fixed explicit λ . Figure 8 makes the comparison concrete, showing that DARP attains sharper low-pass filtering while dispensing entirely with hyperparameter tuning. The learned, difference-conditioned predictor and optional parametric aggregator of the practical algorithm generalize this construction beyond the averaging case, trading the exact spectral equivalence for the additional expressivity that our experiments show to be beneficial.

A.2 ADDITIONAL EXPERIMENTAL DETAILS

A.2.1 RETRIEVAL

The choice of distance function $d(s_q, s_i^*)$ used to retrieve the k neighbors is consequential. While a plain Euclidean distance between states already suffices, our experiments employ a modestly refined

variant that exploits the sequential structure of the data by incorporating trajectory history into the distance computation.

Consider a query trajectory $S_q = (\dots, s_{q,-1}, s_{q,0})$ whose most recent element $s_{q,0}$ is the current query state s_q , and suppose we wish to compute $d(s_q, s_i^*)$ for an expert state s_i^* . We first identify the source trajectory S_j^* from which s_i^* originates, together with its index t within that trajectory, so that s_i^* may be written as $s_{j,t}^*$. Given a lookback parameter ℓ specifying how many past states to consider, the history-augmented distance is

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\|$$

that is, the accumulated Euclidean distance between the current and preceding $\ell - 1$ states of the query and source trajectories, taken over valid indices. Because recent states should exert greater influence on neighbor selection, we down-weight older observations by an exponential decay rate $\gamma \geq 0$, yielding

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\| \cdot e^{-\gamma n}$$

Figure 9 reports how the success rate varies as each of these retrieval parameters—the neighbor count k , the lookback length ℓ , and the decay rate γ —is swept in turn. In every experiment these hyperparameters, along with the retrieval metric and training epoch, are selected by offline validation loss rather than by test performance (Table 11), so no test-time information enters the tuning procedure.

A.2.2 CAN DARP HANDLE TASKS REQUIRING THE REPRESENTATION OF MULTI-MODAL ACTION DISTRIBUTIONS?

We probe DARP’s capacity to represent complex action distributions on the Push-T task of Chi et al. (2024), which demands genuinely multimodal action modeling (Figure 10). Here DARP instantiates its aggregation operator as a Set Transformer head that predicts the parameters of a Gaussian mixture model. Equipped with this GMM head, DARP captures the multimodal action distribution effectively, improving over BC by 22 points on Push-T (Q2; Table 4). This confirms that DARP adapts to multimodal action-distribution modeling requirements through a single, localized change—the choice of a more expressive aggregation head—while leaving the retrieval and difference-conditioning machinery untouched.

A.2.3 CAN DARP HANDLE DISCONTINUOUS ENVIRONMENTS WHERE NEARBY STATES MAY REQUIRE OPPOSING ACTIONS?

A natural concern for any neighbor-based method—and one raised directly by the smoothing perspective developed in the main paper—is its behavior in environments with strong discontinuities, where states that are close in Euclidean distance may nonetheless demand drastically different actions, and where indiscriminate local smoothing could in principle be harmful. We confront this concern directly with a stress test built on a modified version of D4RL’s Umaze environment (Figure 11), whose walls make spatial proximity a deliberately unreliable proxy for action similarity.

Even in this deliberately adversarial discontinuous environment, DARP attains a 57% success rate against BC’s 25% (Q3; Table 5)—more than doubling BC’s performance. We attribute this to the difference vectors and permutation-invariant aggregation, which together let the model discriminate between appropriate and inappropriate neighbors even when spatial proximity does not entail action similarity: conditioning each candidate on its query-to-neighbor difference allows the network to discount neighbors that lie across a discontinuity rather than averaging over them blindly. This is precisely the failure mode such environments are designed to expose, and it delineates the conditions under which retrieval-based smoothing remains appropriate.

A.2.4 CAN DARP RECOVER FROM BC ERROR?

To probe DARP’s robustness to accumulated error directly, we roll out a BC agent in an environment where it is known to fail and, every k steps, fork the environment and launch a DARP agent from the

cloned state. As Fig. 12 shows, even as BC drifts away from the support of the expert demonstrations toward failure, DARP recovers from these same states and scores highly. This demonstrates that DARP is substantially more robust to error accumulation and performs reliably in exactly the mildly out-of-distribution states into which a failing BC agent drifts, complementing the divergence analysis of the main text with an intervention-style test of recovery.

A.2.5 COMPARISON WITH CCIL

Environment	Method	Improvement over BC (%)
Hopper	CCIL	32.6%
	DARP	53.2%
Walker	CCIL	84.1%
	DARP	84.1%
Ant	CCIL	12.6%
	DARP	84.5%
HalfCheetah	CCIL	5.4%
	DARP	418.7%

Table 6: **Relative improvement over BC: DARP versus CCIL.** Percentage improvement over BC on the four standard MuJoCo benchmarks. DARP matches or exceeds CCIL’s relative gains on every task, with a decisive margin on HalfCheetah. Because the two methods are normalized against their respective BC baselines, the comparison isolates the relative benefit each smoothing mechanism confers.

Table 6 compares DARP against CCIL, a baseline that induces smoothness explicitly through a Lipschitz-constrained dynamics model. Using the scores reported in the CCIL paper and measuring percentage improvement over BC, DARP outperforms CCIL on three of the four environments and ties it on the fourth (Walker, 84.1% each), with a striking margin on HalfCheetah (418.7% versus 5.4%). DARP attains these gains through implicit architectural smoothing rather than the explicit Lipschitz constraints CCIL relies on, and thus without the additional smoothness hyperparameters that such constraints require.

A.2.6 DISTANCE METRIC SENSITIVITY

Method	Success Rate (%)
BC	47
DARP (Cosine Similarity)	70
DARP (Euclidean Distance)	75

Table 7: **Distance metric sensitivity.** Success rates using R3M features on the Robosuite Stack task. DARP performs comparably under Euclidean distance and cosine similarity, exceeding BC by 28 and 23 points respectively, which demonstrates robustness to the choice of retrieval metric.

Although every experiment in the paper uses Euclidean distance for neighbor retrieval, alternative metrics such as cosine similarity are natural to consider, particularly in high-dimensional embeddings such as R3M. As Table 7 shows, DARP performs nearly identically under cosine similarity—within 5 points of Euclidean distance and well above BC in both cases—confirming that its gains do not hinge on a particular choice of distance metric.

A.2.7 DARP IN COMBINATION WITH DIFFUSION POLICY

While the experiments in the main text use an MLP backbone, diffusion policy (Chi et al., 2024) has emerged as a state-of-the-art model class for imitation learning, particularly in manipulation. Table 8 shows that DARP is fully compatible with diffusion and, more strikingly, that the two are complementary: pairing the DARP architecture with a diffusion backbone raises success rates over

Method	Stack	Threading	Peg Insertion
BC (MLP)	34	12	28
DARP (MLP)	62	38	40
Diffusion	52	44	54
DARP w/ Diffusion	82	72	66

Table 8: **DARP in combination with a diffusion policy.** DARP delivers substantial improvements atop both MLP and diffusion backbones, and is therefore complementary to, rather than a substitute for, state-of-the-art generative policy classes. Success rates over 50 trials on the state-based Robosuite Stack, Threading, and Peg Insertion tasks. These runs use far fewer demonstrations than Table 2, so the BC and DARP numbers here are not directly comparable to that table; comparisons should be read within each row block (MLP vs. DARP-MLP, and Diffusion vs. DARP-Diffusion).

the MLP-backed DARP by 20, 34, and 26 points on the three tasks, and over standard MLP BC by 48, 60, and 38 points respectively. DARP thus operates as a drop-in enhancement that transfers across backbones rather than competing with them, and the accompanying runtime comparison (Table 9) shows that the MLP variant retains most of the benefit at a fraction of the cost.

Method	Training (s/epoch)	Inference (s/step)
Diffusion	5.610	0.15700
DARP (MLP)	0.559	0.00437

Table 9: **Computational efficiency of DARP (MLP) versus diffusion policy.** Per-epoch training and per-step inference times for the two policy classes. Diffusion policy is markedly more expensive, running $\approx 10\times$ slower in training and $\approx 36\times$ slower at inference, so the MLP-backed DARP retains a decisive efficiency advantage.

The MLP-backed DARP is, moreover, far faster than standard diffusion—especially at inference, as quantified in Table 9. The accuracy gains of the diffusion backbone therefore come at a substantial computational cost that the MLP variant avoids, making the MLP-backed DARP the preferable choice when inference latency is at a premium and the diffusion backbone the preferable choice when peak accuracy is paramount.

A.2.8 DARP TRAINED ON HUMAN DEMONSTRATIONS

Method	Success Rate (%)
BC	45
DARP	60

Table 10: **Results with human demonstrations.** Success rates on the Robosuite Stack task when training on human-collected data in place of data generated by RL policies. DARP retains a clear advantage over BC even under this noisier, less consistent demonstration source.

Because the expert demonstrations for several MuJoCo and Robosuite experiments are collected by a near-optimal reinforcement learning policy, the resulting data may be cleaner and more internally consistent than human demonstrations; it is therefore important to verify that DARP’s advantage is not an artifact of this cleanliness. It is not: when trained on human demonstrations for the Robosuite Stack task, DARP outperforms standard BC by 15 points (60 vs. 45; Table 10), confirming that the method’s benefits carry over to the noisier, less consistent demonstrations characteristic of human data collection.

A.2.9 CHOICE OF RETRIEVAL HYPERPARAMETERS

Environment	Method	Score (Mean)
Hopper	BC	507.49
	DARP	805.69
Stack	BC	0.12
	DARP	0.31

Table 11: **Performance when validation loss selects the training epoch and retrieval hyperparameters.** Mean scores on the Hopper and Stack environments under a fully automatic, offline validation-driven selection procedure. DARP retains a clear advantage over BC without any recourse to test-time information.

Although Figure 9 shows that adversarial retrieval settings—for instance, an extremely small number of neighbors—can drive DARP below standard BC, such settings are readily avoided in practice. Selecting the retrieval hyperparameters (k , lookback ℓ , decay γ , and metric) and training epoch that minimize offline validation loss reliably recovers performant configurations, as Table 11 confirms. DARP therefore realizes its gains under an automatic, validation-driven protocol, without privileged access to test rollouts or test performance.

A.3 PSEUDOCODE

Algorithm 1 provides complete pseudocode for the DARP training procedure, making explicit the neighbor-retrieval, difference-conditioned prediction, and permutation-invariant aggregation steps and their sequence within a single training iteration.

A.4 RUNTIME ANALYSIS

Environment	k	Train (s/epoch)	Test (s/step)
Hopper (1,000 datapoints)	BC	0.102	0.00127
	100	0.126	0.00413
	200	0.128	0.00418
	300	0.130	0.00420
	400	0.131	0.00421
	500	0.130	0.00419
Stack (4,200 datapoints)	BC	0.431	0.00139
	100	0.556	0.00437
	250	0.539	0.00442
	500	0.559	0.00437
	750	0.576	0.00433
	1000	0.591	0.00478
	1500	0.682	0.00452
	2000	0.758	0.00451

Table 12: **Runtime across environments and neighbor counts.** Per-epoch training and per-step inference times (seconds) for BC and for DARP at varying k on the Hopper and Stack datasets. Costs grow only sub-linearly in k , keeping inference well within real-time budgets.

Table 12 quantifies the computational overhead of DARP at both training and inference. The measurements show that cost scales sub-linearly in k and that inference remains comfortably within real-time budgets for robotic control: on the Robosuite Stack task at $k = 500$, inference takes roughly 0.00437 seconds per step on our hardware, corresponding to a control frequency above 230 Hz. The variance-reduction benefits of DARP therefore come at negligible runtime cost on the dataset scales studied here (up to a few thousand states). At substantially larger scales, exact k -NN retrieval would

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Algorithm 1 Difference-Aware Retrieval Policies

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1: Input: Expert demonstrations  $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ , number of neighbors  $k$ 
2: Initialize:  $f$  parameters  $\theta$ 
3: if  $g$  is parametric then
4:   Initialize:  $g$  parameters  $\psi$ 
5: end if
6: // Training Loop
7: while not converged do
8:   Sample batch of query data  $(s_q^*, a_q^*) \sim \mathcal{D}^*$ 
9:   for each query pair  $(s_q^*, a_q^*)$  in batch do
10:    // Find  $k$ -Nearest Neighbors from the entire dataset  $\mathcal{D}^*$ 
11:     $\mathcal{N}_k(s_q^*) \leftarrow \text{argmin-}k_j d(s_q^*, s_j^*)$ 
12:    // Compute Neighbor-based Predictions
13:    for each neighbor index  $i \in \mathcal{N}_k(s_q^*)$  do
14:       $a'_i \leftarrow f_\theta(s_i^*, a_i^*, s_i^* - s_q^*)$ 
15:    end for
16:    // Aggregate Predictions
17:    if  $g$  is parametric then
18:       $\hat{a}_q \leftarrow g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
19:    else
20:       $\hat{a}_q \leftarrow g(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
21:    end if
22:  end for
23:  // Update Parameters based on the batch loss
24:   $\mathcal{L} \leftarrow \sum_{(s_q^*, a_q^*) \in \text{batch}} \|\hat{a}_q - a_q^*\|^2$ 
25:  // Gradient descent step
26:   $\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}$ 
27:  if  $g$  is parametric then
28:     $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}$ 
29:  end if
30: end while
31: Output: Trained parameters  $\theta$  and, if applicable,  $\psi$ 

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1350 become the dominant cost, and approximate nearest-neighbor indexing would be the natural remedy;
1351 characterizing DARP under such indexing, and on datasets of hundreds of thousands to millions of
1352 states, is a promising direction that our present experiments do not settle.
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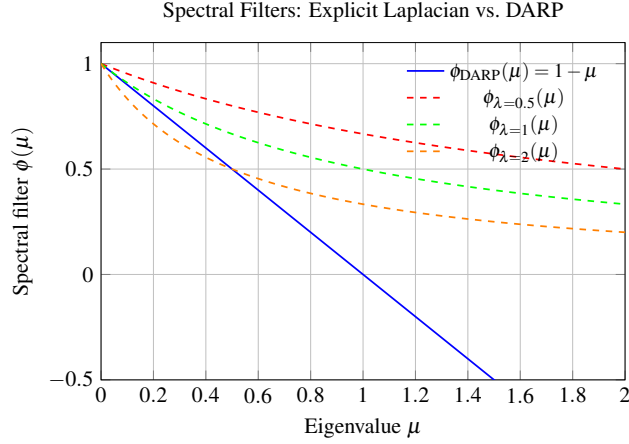


Figure 8: DARP achieves sharper low-pass filtering than explicit Laplacian regularization for any fixed λ .

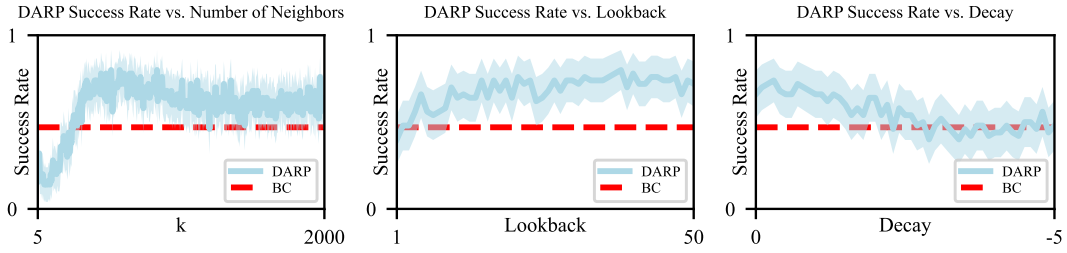


Figure 9: DARP performance as the three retrieval hyperparameters are swept. **(left)** As a function of the number of retrieved neighbors k , performance is poor for very small k , attains a global optimum near $k \approx 500$, and plateaus just above BC’s success rate as k approaches the dataset size. **(center)** Performance improves monotonically as more trajectory history is incorporated into the distance computation, dropping below BC only when little or no history is used. **(right)** Performance is sensitive to how strongly older observations are down-weighted: an excessively large decay rate effectively discards history, collapsing DARP to the low-lookback regime and again falling below BC. Success rate is measured over 50 trials on the Robosuite Stack environment, with 95% confidence intervals. These curves motivate the validation-loss selection protocol of Table 11, which recovers performant settings automatically.

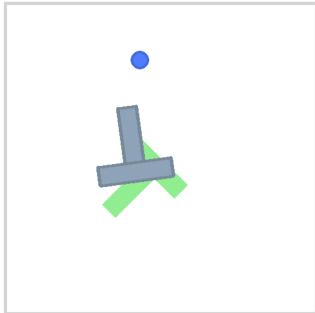


Figure 10: **Push-T Environment.** The agent controls the blue circle to push the T-shaped block onto a target, a task whose solution space is genuinely multimodal.

Method	Score
BC	48 ± 8
DARP	70 ± 8

Table 4: **Push-T Results.** Averaged over 100 trials, DARP outperforms BC by 22 points (70 vs. 48) on this multimodal task.

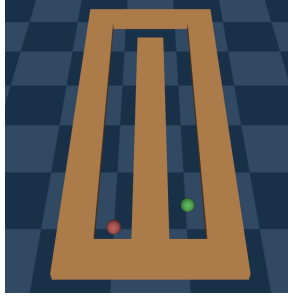


Figure 11: **Long maze environment.** A force-actuated ball must be driven from the green start to the red destination through a layout in which spatially adjacent states, separated by walls, frequently require opposing actions.

Method	Succ. (%)
BC	25
DARP	57

Table 5: **Long maze results.** Averaged over 100 trials, DARP more than doubles BC’s success rate in this discontinuous environment.

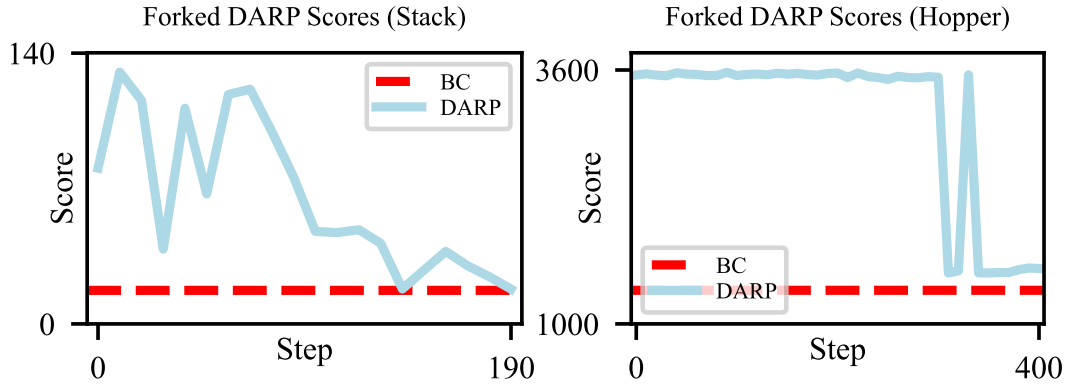


Figure 12: On two tasks (Robosuite Stack and MuJoCo Hopper), we roll out a BC agent and fork the environment every k steps (here $k = 10$), launching a DARP agent from each fork. Even as BC nears the end of its failing rollout, DARP continues to score highly from these forked states, ceasing to do so only about halfway through the Stack rollout and roughly 80% of the way through the Hopper rollout—evidence that DARP recovers from states into which a failing BC agent has already drifted, up to the point at which the state leaves the demonstration support entirely.